KENNESAW STATE
UNIVERSITY

# CS 4277
# Preliminaries and Data Manipulation

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# Tools

- Python
  - Pip
  - Conda
- Jupyter notebook
  - Much easier to keep track of your experiments
  - Run codes from different blocks
  - Obviously you should put longer code into modules
- Reveal (for notebook slides)
  - conda install –c conda-forge rise

# Laptop/Desktop/Generic Cloud with Linux

- Conda: conda

Windows macOS Linux

These four commands quickly and quietly install the latest 64-bit version of the installer and then clean up after themselves. To install a different version or architecture of Miniconda for Linux, change the name of the `.sh` installer in the `wget` command.

```
mkdir -p ~/miniconda3
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O ~/miniconda3/miniconda.sh
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3
rm -rf ~/miniconda3/miniconda.sh
```

- Create environment: conda
- GPU support
  - Install NVIDIA drivers/CUDA/CUDNN/TensorRT

# Laptop / Desktop with MacOS

- Conda: conda

Windows macOS Linux

These four commands quickly and quietly install the latest M1 macOS version of the installer and then clean up after themselves. To install a different version or architecture of Miniconda for macOS, change the name of the `.sh` installer in the `curl` command.

```
mkdir -p ~/miniconda3
curl https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-arm64.sh -o ~/miniconda3/miniconda.sh
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3
rm -rf ~/miniconda3/miniconda.sh
```

- Create environment: conda
- GPU support
  - Install NVIDIA drivers/CUDA/CUDNN/TensorRT

# Preliminaries

- All machine learning is concerned with extracting information from data
  - Practical skills needed for **storing**, **manipulating**, and **preprocessing** data
  - Scientific Python: Numpy, Pandas, Scipy, Scikit-learn, Scikit-image, OpenCV
  - d2l
  - d2l
- Machine learning typically requires working with large datasets,
  - Datasets considers as tables
    - rows correspond to examples and
    - columns correspond to attributes.
- Linear algebra
  - set of techniques for working with tabular data
- Deep learning is all about optimization
  - model with some parameters
  - find those that fit our data *the best*

# Data Manipulation

- Two important steps
  - acquire them
  - process them
- $n$-dimensional array, which is also called the *tensor*
- *tensor class (*Tensor in both PyTorch and TensorFlow) is similar to NumPy's ndarray.
- A tensor represents a (possibly multi-dimensional) array of numerical values.
- Two benefits of tensors:
  - GPU is well-supported to accelerate the computation
  - tensor class supports automatic differentiation

# Linear Algebra

# Scalars

- Simple operations
  - c = a + b
    c = a · b
    c = sin a
- Length

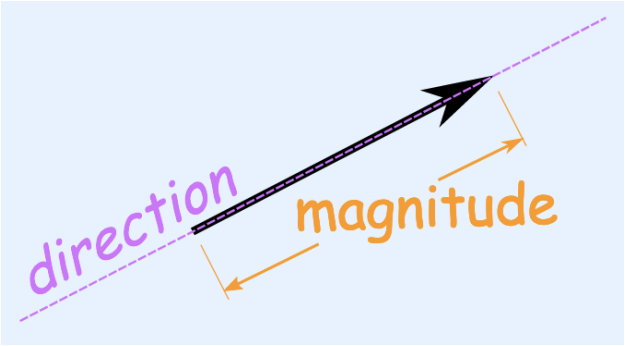
$$|a| = \begin{cases} a & \text{if} a > 0 \\ -a & \text{otherwise} \end{cases}$$

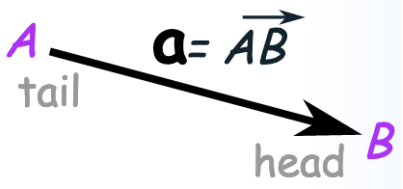
$$|a + b| \leq |a| + |b|$$

$$|a \cdot b| = |a| \cdot |b|$$

# Vectors

- **A vector has magnitude and direction**
- Displacement, velocity, acceleration, force and momentum are all vectors.
- **Distance vs Displacement**
  - Distance is a scalar ("3 km")
  - Displacement is a vector ("3 km Southeast")



# Vectors

- **Simple operations**

$$c = a + b \quad \text{where } c_i = a_i + b_i$$
$$c = \alpha \cdot b \quad \text{where } c_i = \alpha b_i$$
$$c = \sin a \quad \text{where } c_i = \sin a_i$$

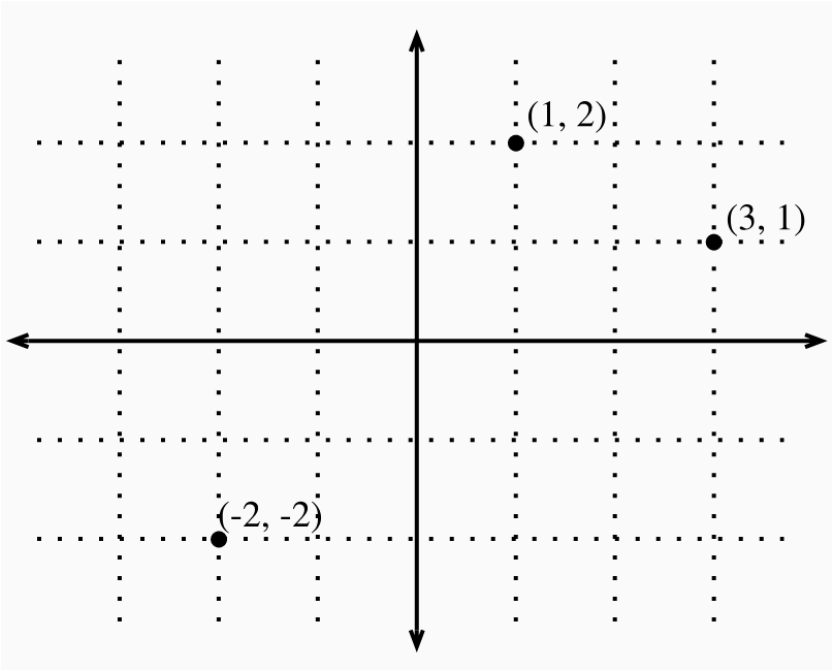
Definition of a vector space

- **Length**

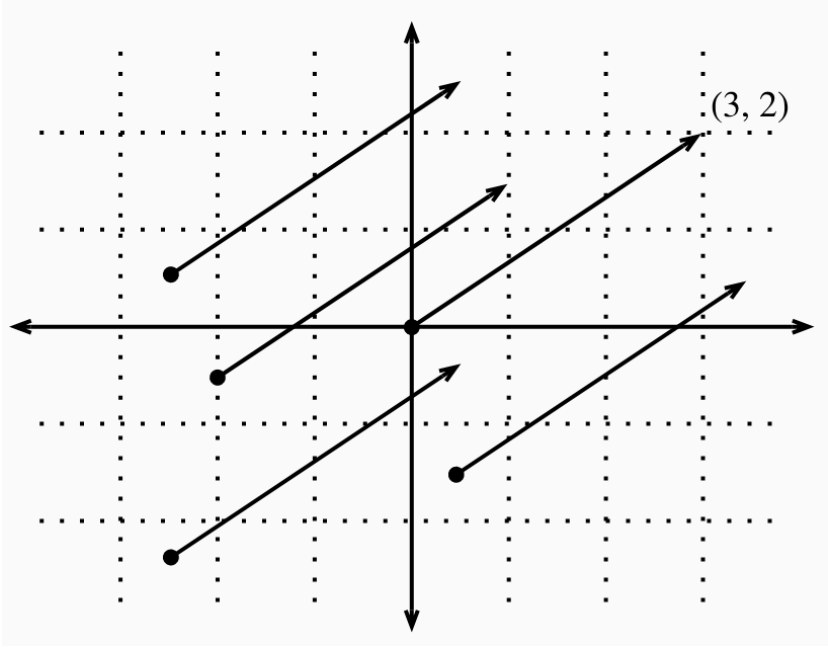
$$\|a\|_2 = \left[\sum_{i=1}^{m} a_i^2\right]^{\frac{1}{2}}$$
$$\|a\| \geq 0 \text{ for all } a$$
$$\|a + b\| \leq \|a\| + \|b\|$$

Definition of a norm

# Vectors




vectors as points in the plane




vectors: as directions in space

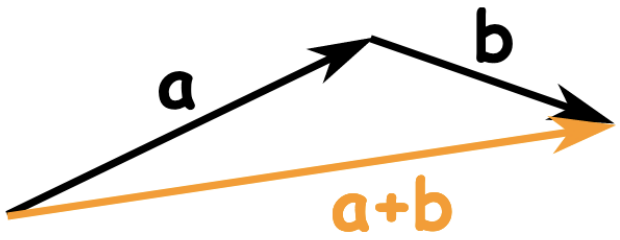
every vector drawn is a representation of the vector $(3,2)^\top$

a single vector's default orientation is a column vector $\mathbf{x} = \begin{bmatrix} 1 \\ 7 \\ 0 \\ 1 \end{bmatrix}$

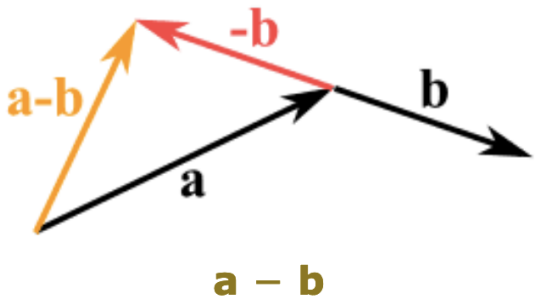
$$\mathbf{x}^\top = \begin{bmatrix} 1 & 7 & 0 & 1 \end{bmatrix}$$

# Vectors

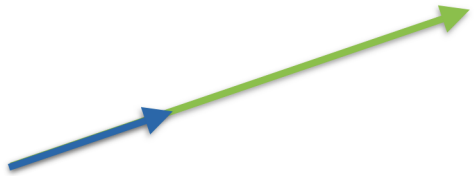
- add two vectors by joining them head-to-tail




- subtract one vector from another

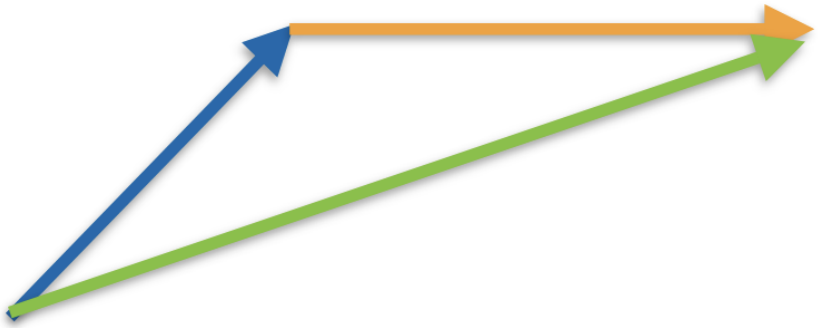



- multiply two vectors
  - The scalar or Dot Product (the result is a scalar).
  - The vector or Cross Product (the result is a vector).

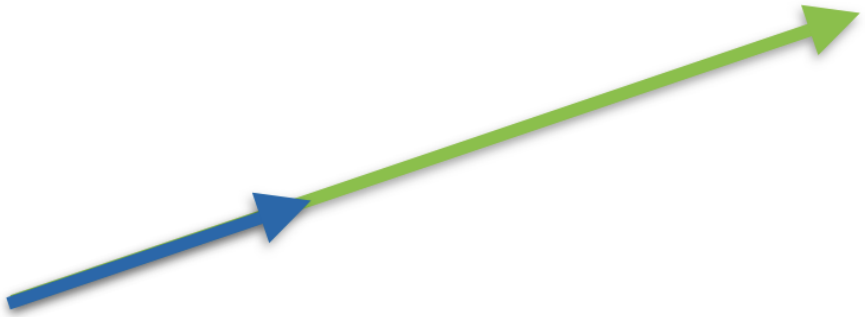


$$c = \alpha \cdot b$$

# Vectors



$$c = a + b$$



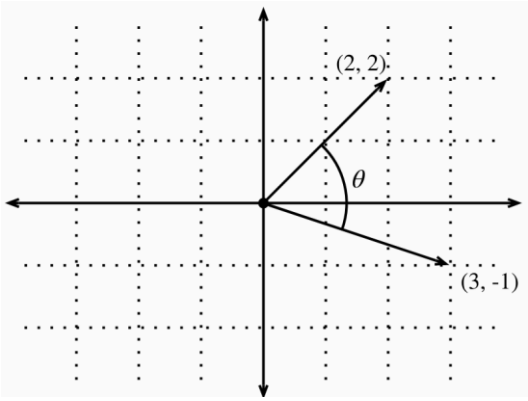
$$c = \alpha \cdot b$$

single vector’s default orientation is a column vector,

# Vectors

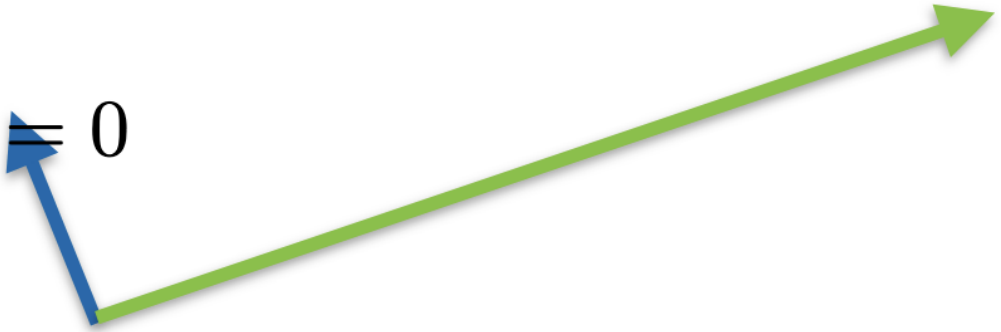
- **Dot product**

$$a^\top b = \sum_i a_i b_i$$




- **Orthogonality**

$$a^\top b = \sum_i a_i b_i = 0$$



(e.g. if we have two vectors that are orthogonal with a third, their linear combination is it, too)

# Cosine Similarity

$$\cos(\theta) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|}$$

- The cosine takes a maximum value of 1 when the two vectors point in the same direction, a minimum value of −1 when they point in opposite directions, and a value of 0 when the two vectors are orthogonal.

# Matrices

- A matrix is an array of numbers (one or more rows, one or more columns).
- Simple operations

$$C = A + B \qquad \text{where } C_{ij} = A_{ij} + B_{ij}$$

$$C = \alpha \cdot B \qquad \text{where } C_{ij} = \alpha B_{ij}$$

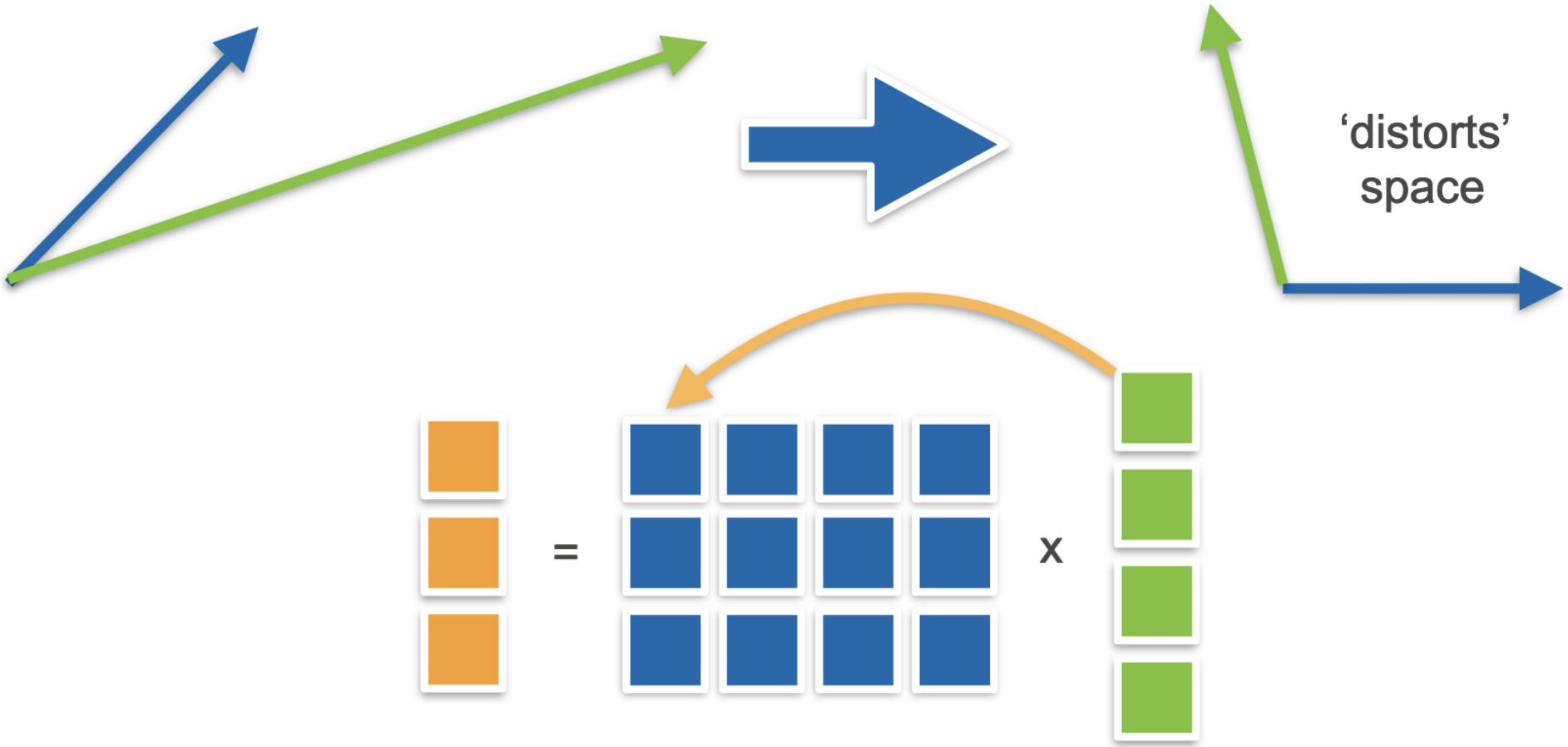
$$C = \sin A \qquad \text{where } C_{ij} = \sin A_{ij}$$

# Matrices

- **Multiplications (matrix vector)**

$$c = Ab \text{ where } c_i = \sum_j A_{ij} b_j$$

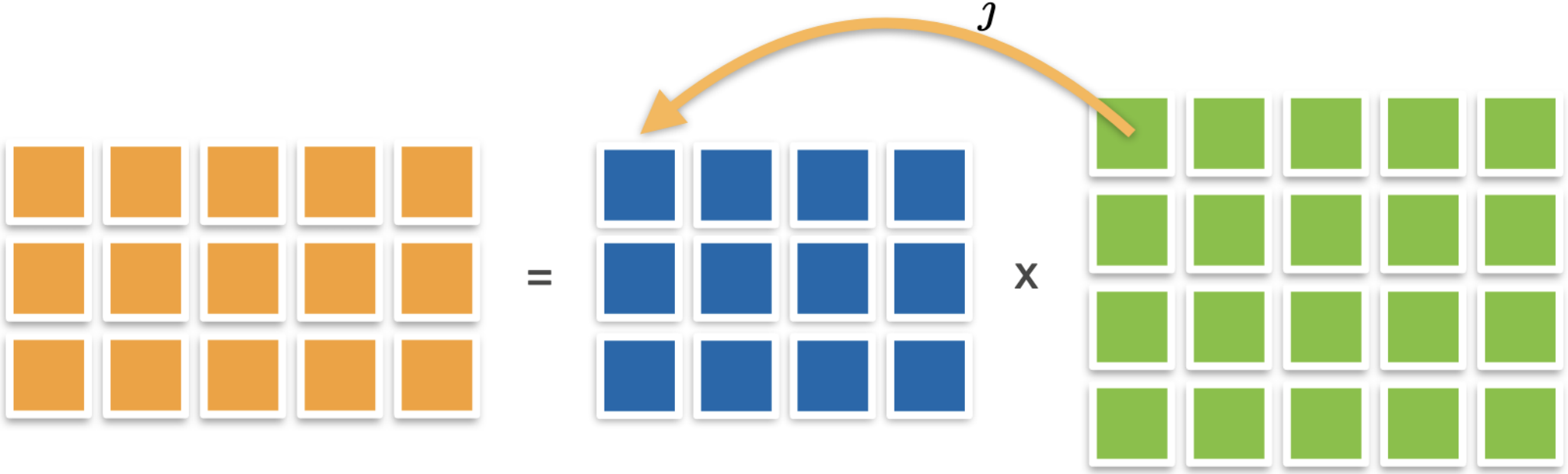
# Matrices



# Matrices

- **Multiplications (matrix matrix)**

$$C = AB \text{ where } C_{ik} = \sum_j A_{ij} B_{jk}$$



# Rank

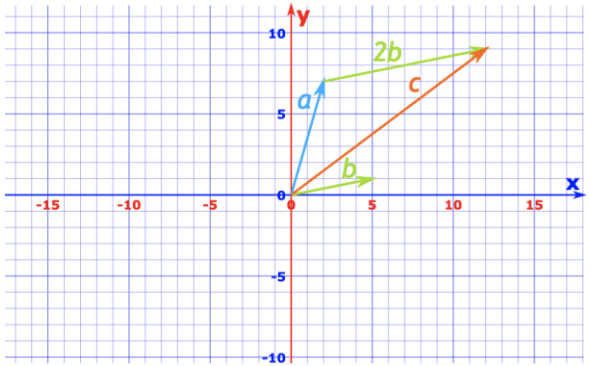
- *Given* $n \times m$ matrix,
  - what dimension space the matrix maps into.
  - Answer: Rank
- linear dependence - compression of space into a lower dimension
- Rank: rank of a matrix $\mathbf{A}$ is the largest number of linearly independent columns amongst all subsets of columns

$$\mathbf{B} = \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}$$

Rank(B)=1

# Linear Dependence

- Linear means we can multiply by a constant, but no powers or other functions. The constant can be any real number (0, 1, any whole number, fraction, negatives, etc.).
- Dependence means they depend on each other; in other words, we can add some up (after multiplying by a constant) to make another one.




**c** = **a** + 2**b**,
so **c** is **linearly dependent** on **a** and **b**

# Geometry of Linear Transformations

Suppose that we have some matrix:

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

If we want to apply this to an arbitrary vector $\mathbf{v} = [x, y]^\top$, we multiply and see that

$$\begin{aligned} \mathbf{A}\mathbf{v} &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix} \\ &= x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix} \\ &= x \left\{ \mathbf{A} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} + y \left\{ \mathbf{A} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}. \end{aligned}$$

- a matrix transforms any vector in terms of how it transforms two specific vectors: [1,0]T and [0,1]T

# Linear Equations Using Matrices

$$x + y + z = 6$$
$$2y + 5z = -4$$
$$2x + 5y - z = 27$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 5 \\ 2 & 5 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -4 \\ 27 \end{bmatrix}$$

"Dot Product"

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 5 \\ 2 & 5 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x + y + z \\ 2y + 5z \\ 2x + 5y - z \end{bmatrix}$$

$AX = B$

$X = A^{-1}B$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 5 \\ 2 & 5 & -1 \end{bmatrix}^{-1} = \frac{1}{-21} \begin{bmatrix} -27 & 6 & 3 \\ 10 & -3 & -5 \\ -4 & -3 & 2 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{-21} \begin{bmatrix} -27 & 6 & 3 \\ 10 & -3 & -5 \\ -4 & -3 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ -4 \\ 27 \end{bmatrix} = \frac{1}{-21} \begin{bmatrix} -105 \\ -63 \\ 42 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$$

# Invertibility

- multiplication by a matrix with linearly dependent columns cannot be undone
- multiplication by a full-rank matrix (i.e., some $\mathbf{A}$ that is $n \times n$ matrix with rank $n$), we should always be able to undo it.
- identity matrix which leaves our data unchanged when applied.

# Invertibility

The inverse of $A$ is $A^{-1}$ only when:

$$AA^{-1} = A^{-1}A = \mathbf{I}$$

Sometimes there is no inverse at all.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

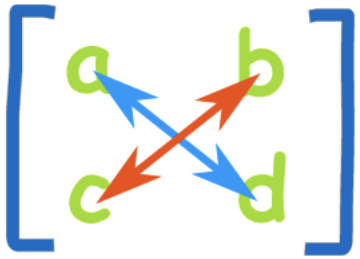
In other words: **swap** the positions of a and d, put **negatives** in front of b and c, and **divide** everything by **ad−bc** .

**ad−bc** is called the determinant

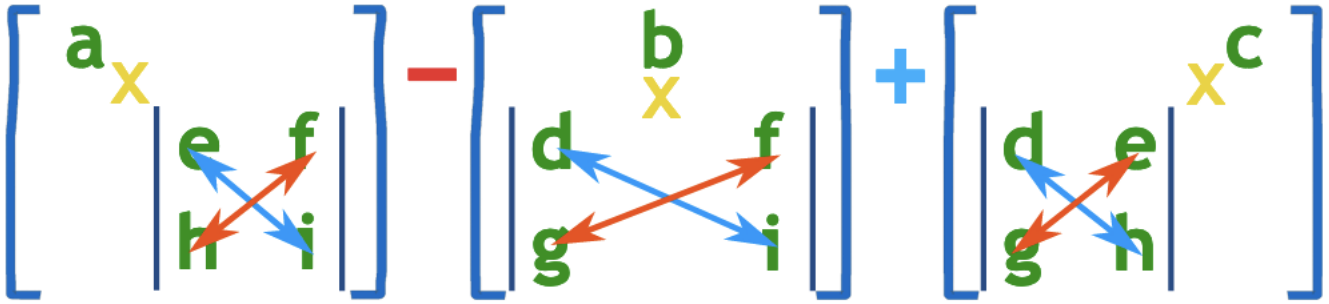
# Determinant

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|A| = ad - bc$$




$$|A| = a(ei - fh) - b(di - fg) + c(dh - eg)$$




$$|A| = a \cdot \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \cdot \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \cdot \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

https://www.mathsisfun.com/algebra/matrix-determinant.html

# Why Do We Need an Inverse?

- matrices we **don't divide**! Instead, **multiply by an inverse**
- we want to find matrix X, and we know matrix A and B
  - $XA = B$
  - $XAA^{-1} = BA^{-1}$ (we can not divide)
  - $XI = BA^{-1}$
  - $X = BA^{-1}$

# A Real-Life Example: Bus and Train

- A group took a trip on a **bus**, at $3 per child and $3.20 per adult for a total of $118.40.
- They took the **train** back at $3.50 per child and $3.60 per adult for a total of $135.20.
- How many children, and how many adults?

$$\begin{array}{cc} \text{Child} & \text{Adult} \\ \end{array} \quad \begin{array}{cc} \text{Bus} & \text{Train} \\ \end{array} \qquad \begin{array}{cc} \text{Bus} & \text{Train} \\ \end{array}$$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & 3.5 \\ 3.2 & 3.6 \end{bmatrix} = \begin{bmatrix} 118.4 & 135.2 \end{bmatrix}$$

# A Real-Life Example: Bus and Train

$$\begin{bmatrix} 3 & 3.5 \\ 3.2 & 3.6 \end{bmatrix}^{-1} = \frac{1}{3\times3.6-3.5\times3.2}\begin{bmatrix} 3.6 & -3.5 \\ -3.2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -9 & 8.75 \\ 8 & -7.5 \end{bmatrix}$$

$$X = BA^{-1}$$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} 118.4 & 135.2 \end{bmatrix}\begin{bmatrix} -9 & 8.75 \\ 8 & -7.5 \end{bmatrix}$$

$$= \begin{bmatrix} 118.4\times-9 + 135.2\times8 & 118.4\times8.75 + 135.2\times-7.5 \end{bmatrix}$$

$$= \begin{bmatrix} 16 & 22 \end{bmatrix}$$

# Eigenvector and Eigenvalue

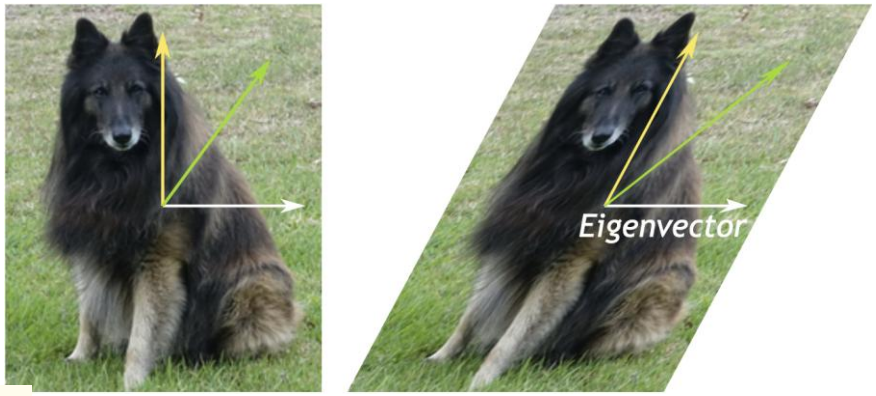
- eigenvector **does not change direction** in a transformation
- For a square matrix **A**, an Eigenvector and Eigenvalue make this equation true:

$$Av = \lambda v$$

Matrix

Eigenvector

Eigenvalue




- **eigenvalue of 6**

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix}$$

Matrix, A

$$\begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Eigen Vector

# Eigenvector and Eigenvalue

$Av$ gives us:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -6\times1+3\times4 \\ 4\times1+5\times4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

$\lambda v$ gives us :

$$6\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

Yes they are equal! So $Av = \lambda v$ as promised.

# How do we find these eigen things?

We start by finding the **eigenvalue**: we know this equation must be true:

$$Av = \lambda v$$

Now let us put in an identity matrix so we are dealing with matrix-vs-matrix:

$$Av = \lambda Iv$$

Bring all to left hand side:

$$Av - \lambda Iv = 0$$

If **v** is non-zero then we can solve for $\lambda$ using just the determinant:

$$| A - \lambda I | = 0$$

# Solve for λ

Start with $|A - \lambda I| = 0$

$$\left| \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

Which is:

$$\begin{vmatrix} -6-\lambda & 3 \\ 4 & 5-\lambda \end{vmatrix} = 0$$

Calculating that determinant gets:

$$(-6-\lambda)(5-\lambda) - 3\times4 = 0$$

Which then gets us this Quadratic Equation:

$$\lambda^2 + \lambda - 42 = 0$$

And solving it gets:

$$\lambda = -7 \text{ or } 6$$

And yes, there are **two** possible eigenvalues.

# Example (continued)

Start with:

$$Av = \lambda v$$

Put in the values we know:

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 6 \begin{bmatrix} x \\ y \end{bmatrix}$$

After multiplying we get these two equations:

$$\begin{aligned} -6x + 3y &= 6x \\ 4x + 5y &= 6y \end{aligned}$$

Bringing all to left hand side:

$$\begin{aligned} -12x + 3y &= 0 \\ 4x - 1y &= 0 \end{aligned}$$

Eigen is a German word meaning "own" or "typical"
*"das ist ihnen **eigen**"* is German for *"that is **typical** of them"*

# Example: find the eigenvalues for this 3x3 matrix

Now the determinant should equal zero:

$$\begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & 4-\lambda & 5 \\ 0 & 4 & 3-\lambda \end{vmatrix} = 0$$

Which is:

$$(2-\lambda)\ [\ (4-\lambda)(3-\lambda) - 5\times 4\ ] = 0$$

This ends up being a cubic equation, but just looking at it here we see one of the roots is **2** (because of $2-\lambda$), and the part inside the square brackets is Quadratic, with roots of **−1** and **8**.

So the Eigenvalues are **−1**, **2** and **8**

# Using Tensorflow or Pytorch

- https://d2l.ai/chapter_appendix-mathematics-for-deep-learning/eigendecomposition.html

# Special Matrices

- **Symmetric, antisymmetric** $A_{ij} = A_{ji}$ and $A_{ij} = -A_{ji}$

- **Positive definite**

$$\|x\|^2 = x^\top x \geq 0 \text{ generalizes to } x^\top A x \geq 0$$

(all positive eigenvalues)

# Special Matrices

- **Orthogonal Matrices**
  - All rows of the matrix are orthogonal to each other
  - All rows of the matrix have unit length

$$U \text{ with } \sum_j U_{ij} U_{kj} = \delta_{ik}$$

Show that $U^\top U = \mathbf{1}$

  - Rewrite in matrix form

$$UU^\top = \mathbf{1}$$

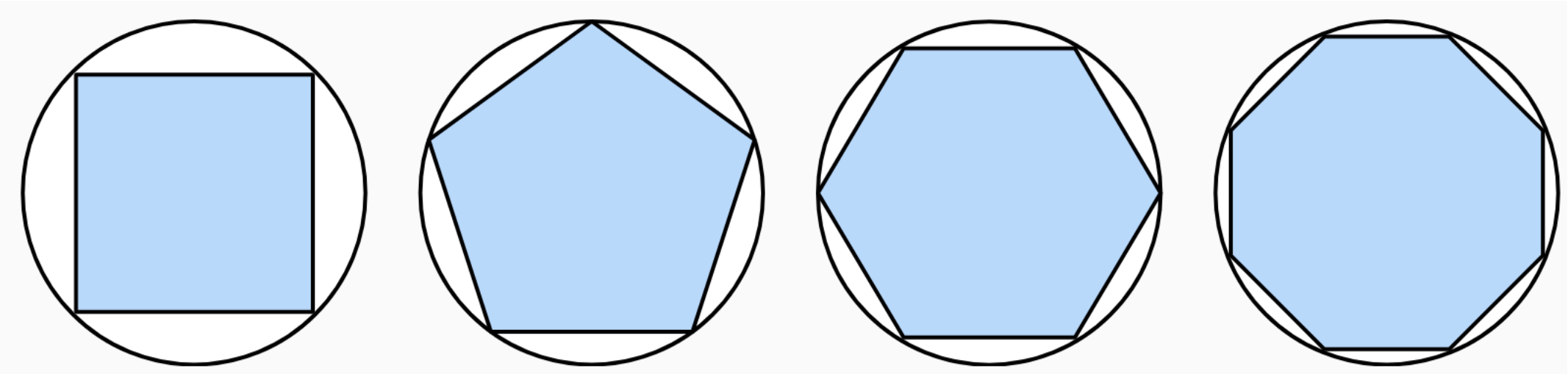
Show that P is orthogonal

- **Permutation Matrices**

$$P \text{ where } P_{ij} = 1 \text{ if and only if } j = \pi(i)$$

# Calculus

# Find Area



Find the area of a circle with the method of exhaustion

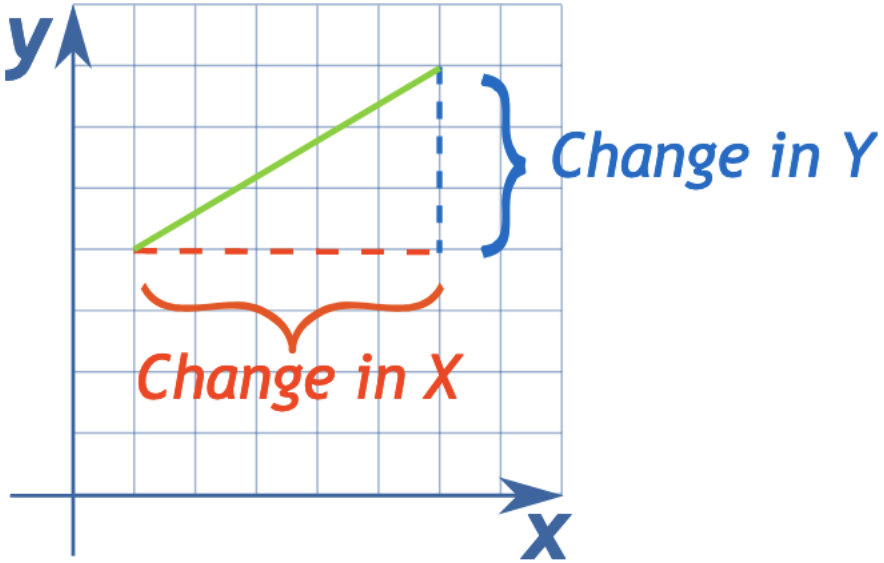
# Derivatives and Differentiation

- derivatives, a crucial step in nearly all deep learning optimization algorithms
- loss functions that are differentiable with respect to our model's parameters
- this means that for each parameter, we can determine how rapidly the loss would increase or decrease were we to increase or decrease that parameter by an infinitesimally small amount

# Derivatives

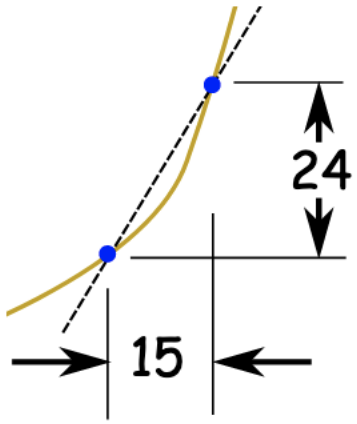
It is all about slope!

$$\text{Slope} = \frac{\text{Change in Y}}{\text{Change in X}}$$



# Derivatives

We can find an **average** slope between two points.

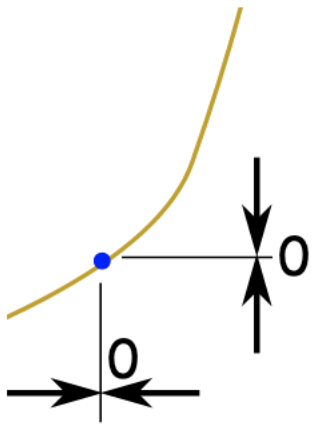



average slope = $\frac{24}{15}$

# Derivatives

But how do we find the slope **at a point**?

There is nothing to measure!

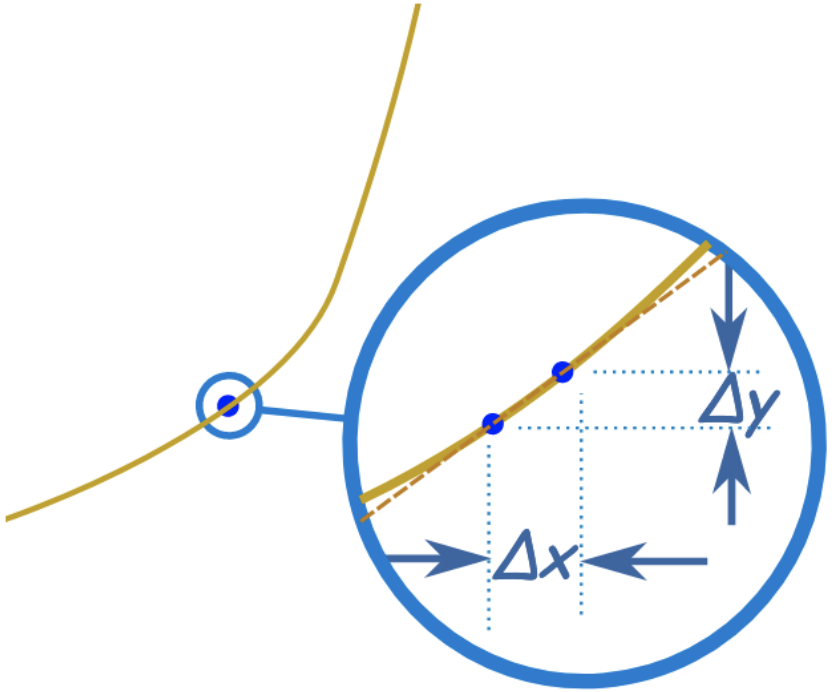



$$\text{slope} = \frac{0}{0} = ???$$

# Derivatives

But with derivatives we use a small difference ...

... then have it **shrink towards zero**.



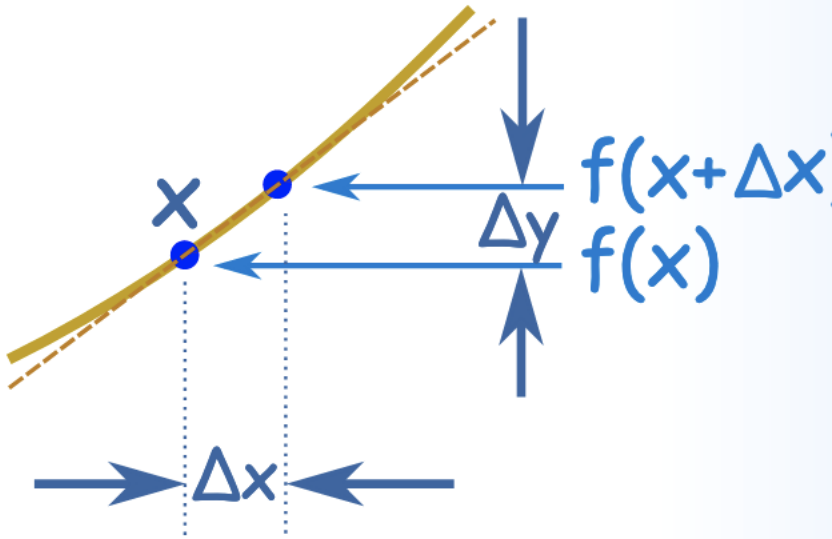
To find the derivative of a function $y = f(x)$ we use the slope formula:

$$\text{Slope} = \frac{\text{Change in Y}}{\text{Change in X}} = \frac{\Delta y}{\Delta x}$$

And (from the diagram) we see that:

x changes from $x$ to $x+\Delta x$

y changes from $f(x)$ to $f(x+\Delta x)$




Now follow these steps:

- Fill in this slope formula: $\frac{\Delta y}{\Delta x} = \frac{f(x+\Delta x) - f(x)}{\Delta x}$
- Simplify it as best we can
- Then make **Δx** shrink towards zero.

# Example: the function $f(x) = x^2$

We know $\mathbf{f(x) = x^2}$, and we can calculate $\mathbf{f(x+\Delta x)}$ :

Start with: $\mathbf{f(x+\Delta x) = (x+\Delta x)^2}$

Expand $(x + \Delta x)^2$: $\mathbf{f(x+\Delta x) = x^2 + 2x\,\Delta x + (\Delta x)^2}$

The slope formula is: $$\frac{f(x+\Delta x) - f(x)}{\Delta x}$$

Put in $\mathbf{f(x+\Delta x)}$ and $\mathbf{f(x)}$: $$\frac{x^2 + 2x\,\Delta x + (\Delta x)^2 - x^2}{\Delta x}$$

Simplify ($x^2$ and $-x^2$ cancel): $$\frac{2x\,\Delta x + (\Delta x)^2}{\Delta x}$$

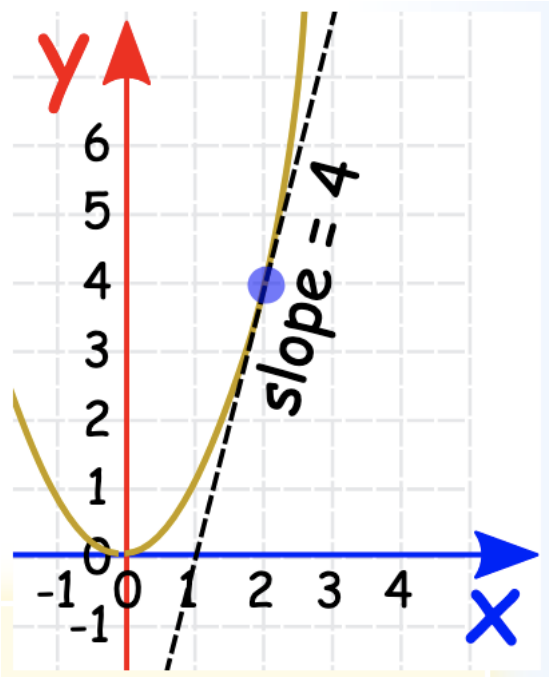
Simplify more (divide through by $\Delta x$): $$= 2x + \Delta x$$

Then, **as $\Delta x$ heads towards 0** we get: $$= 2x$$

Result: the derivative of $\mathbf{x^2}$ is $\mathbf{2x}$

## So what does $\frac{d}{dx}x^2 = 2x$ mean?

- It means that, for the function $x^2$, the slope or "rate of change" at any point is **2x**.
- So when **x=2** the slope is **2x = 4**, as shown here:
- Or when **x=5** the slope is **2x = 10**, and so on.



# Notation

- "Shrink towards zero" is actually written as a limit like this

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

"The derivative of **f** equals
**the limit as Δx goes to zero** of f(x+Δx) - f(x) over Δx"

- sometimes the derivative is written like this

$$\frac{dy}{dx} = \frac{f(x+dx) - f(x)}{dx}$$

# Derivatives

- Suppose that we have a function $f:\mathbb{R}\rightarrow\mathbb{R}$, whose input and output are both scalars. The *derivative* of $f$ is defined as

$$f'(x) = \lim_{h\rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

- The so-called instantaneous rate of change is based on the variation $h$ in $x$, which approaches 0

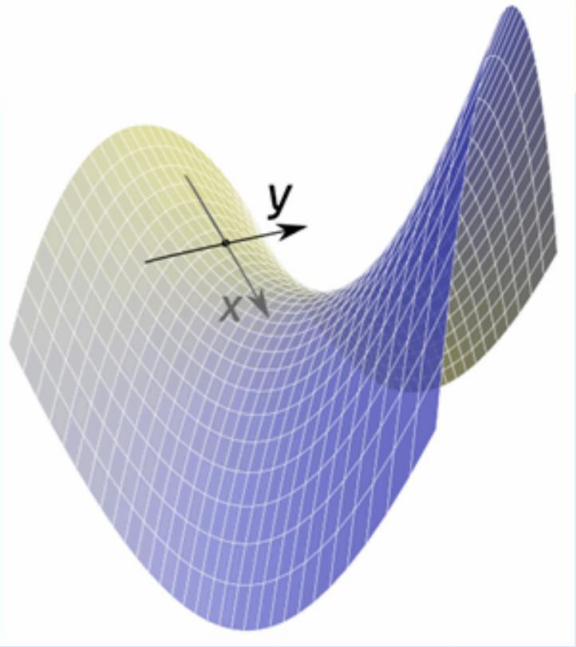
# Partial Derivatives

- A Partial Derivative is a derivative where we hold some variables constant.

Example: a function for a surface that depends on two variables **x** and **y**

When we find the slope in the **x** direction (while keeping **y** fixed) we have found a partial derivative.

Or we can find the slope in the **y** direction (while keeping **x** fixed).



# Partial Derivatives

$f(x, y) = x^2 + y^3$

- its **partial** derivative **with respect to x** when we treat **y as a constant**

$$f'_x = 2x + 0 = 2x$$

  - *the derivative of $x^2$ (with respect to x) is 2x*
  - *we **treat y as a constant**, so $y^3$ is also a constant (imagine y=7, then $7^3$=343 is also a constant), and the derivative of a constant is 0*

- partial derivative **with respect to y**, we treat **x as a constant**

$$f'_y = 0 + 3y^2 = 3y^2$$

  - *we now **treat x as a constant**, so $x^2$ is also a constant, and the derivative of a constant is 0*
  - *the derivative of $y^3$ (with respect to y) is $3y^2$*

# Gradients

- To obtain gradient of Multivariate function concatenate all the partial derivatives
- Input of function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is an $n$-dimensional vector $\mathbf{x}=[x_1, x_2, \ldots, x_n]^\top$ and the output is a scalar.
- The gradient of the function $f(\mathbf{x})$ with respect to $\mathbf{x}$ is a vector of $n$ partial derivatives:

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \left[ \frac{\partial f(\mathbf{x})}{\partial x_1}, \frac{\partial f(\mathbf{x})}{\partial x_2}, \ldots, \frac{\partial f(\mathbf{x})}{\partial x_n} \right]^\top$$

# Gradients Rules

- Let $\mathbf{x}$ be an $n$-dimensional vector, the following rules are often used when differentiating multivariate functions:

  - For all $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\nabla_{\mathbf{x}} \mathbf{A}\mathbf{x} = \mathbf{A}^\top$,
  - For all $\mathbf{A} \in \mathbb{R}^{n \times m}$, $\nabla_{\mathbf{x}} \mathbf{x}^\top \mathbf{A} = \mathbf{A}$,
  - For all $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\nabla_{\mathbf{x}} \mathbf{x}^\top \mathbf{A}\mathbf{x} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$,
  - $\nabla_{\mathbf{x}} \|\mathbf{x}\|^2 = \nabla_{\mathbf{x}} \mathbf{x}^\top \mathbf{x} = 2\mathbf{x}$.

Similarly, for any matrix $\mathbf{X}$, we have $\nabla_{\mathbf{X}} \|\mathbf{X}\|_F^2 = 2\mathbf{X}$.

# Chain Rule

- "Function Composition" is applying one function to the results of another

The result of f() is sent through g()

It is written: $(g \circ f)(x)$

Which means: $g(f(x))$

$$(g \circ f)(x) = (2x+3)^2$$

$$(f \circ g)(x) = 2x^2+3$$

- multivariate functions in deep learning are often *composite*
- functions of a single variable, $y = f(u)$ and $u = g(x)$ are both differentiable

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

# Chain Rule

- functions have an arbitrary of variables
- differentiable function $y$ has variables $u_1, u_2, \ldots, u_m$, where each differentiable function $u_i$ has variables $x_1, x_2, \ldots, x_n$
- $y$ is a function of $x_1, x_2, \ldots, x\text{number}_n$
- chain rule

$$\frac{dy}{dx_i} = \frac{dy}{du_1}\frac{du_1}{dx_i} + \frac{dy}{du_2}\frac{du_2}{dx_i} + \cdots + \frac{dy}{du_m}\frac{du_m}{dx_i}$$

# Chain Rule: Generalize to Vectors

Chain rule for scalars:

$$y = f(u),\ u = g(x) \qquad \frac{\partial y}{\partial x} = \frac{\partial y}{\partial u}\frac{\partial u}{\partial x}$$

Generalize to vectors straightforwardly

$$\frac{\partial y}{\partial \mathbf{x}} = \frac{\partial y}{\partial u}\frac{\partial u}{\partial \mathbf{x}} \qquad \frac{\partial y}{\partial \mathbf{x}} = \frac{\partial y}{\partial \mathbf{u}}\frac{\partial \mathbf{u}}{\partial \mathbf{x}} \qquad \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \frac{\partial \mathbf{y}}{\partial \mathbf{u}}\frac{\partial \mathbf{u}}{\partial \mathbf{x}}$$

$$(1,n) \quad (1,)\ (1,n) \qquad (1,n) \quad (1,k)\ \ (k,n) \qquad (m,n)\ \ (m,k)\ (k,n)$$

# Example 1

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \frac{\partial \mathbf{y}}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \mathbf{x}}$$

Assume $\mathbf{x}, \mathbf{w} \in \mathbb{R}^n, \quad y \in \mathbb{R}$

$$z = \left(\langle \mathbf{x}, \mathbf{w} \rangle - y\right)^2$$

Compute $\dfrac{\partial z}{\partial \mathbf{w}}$

$$\begin{aligned}
\frac{\partial z}{\partial \mathbf{w}} &= \frac{\partial z}{\partial b} \frac{\partial b}{\partial a} \frac{\partial a}{\partial \mathbf{w}} \\
&= \frac{\partial b^2}{\partial b} \frac{\partial a - y}{\partial a} \frac{\partial \langle \mathbf{x}, \mathbf{w} \rangle}{\partial \mathbf{w}} \\
&= 2b \cdot 1 \cdot \mathbf{x}^T \\
&= 2\left(\langle \mathbf{x}, \mathbf{w} \rangle - y\right) \mathbf{x}^T
\end{aligned}$$

Decompose

$$\begin{aligned}
a &= \langle \mathbf{x}, \mathbf{w} \rangle \\
b &= a - y \\
z &= b^2
\end{aligned}$$

# Example 2

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \frac{\partial \mathbf{y}}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \mathbf{x}}$$

Assume $\mathbf{X} \in \mathbb{R}^{m \times n}, \quad \mathbf{w} \in \mathbb{R}^{n}, \quad \mathbf{y} \in \mathbb{R}^{m}$

$$z = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$$

Compute $\dfrac{\partial z}{\partial \mathbf{w}}$

$$\begin{aligned}
\frac{\partial z}{\partial \mathbf{w}} &= \frac{\partial z}{\partial \mathbf{b}} \frac{\partial \mathbf{b}}{\partial \mathbf{a}} \frac{\partial \mathbf{a}}{\partial \mathbf{w}} \\
&= \frac{\partial \|\mathbf{b}\|^2}{\partial \mathbf{b}} \frac{\partial \mathbf{a} - \mathbf{y}}{\partial \mathbf{a}} \frac{\partial \mathbf{X}\mathbf{w}}{\partial \mathbf{w}} \\
&= 2\mathbf{b}^T \times \mathbf{I} \times \mathbf{X} \\
&= 2\left(\mathbf{X}\mathbf{w} - \mathbf{y}\right)^T \mathbf{X}
\end{aligned}$$

Decompose

$$\mathbf{a} = \mathbf{X}\mathbf{w}$$
$$\mathbf{b} = \mathbf{a} - \mathbf{y}$$
$$z = \|\mathbf{b}\|^2$$

# Some Derivative Rules

# Derivative Rules

| Common Functions | Function | Derivative |
|---|---|---|
| Constant | $c$ | $0$ |
| Line | $x$ | $1$ |
| | $ax$ | $a$ |
| Square | $x^2$ | $2x$ |
| Square Root | $\sqrt{x}$ | $(½)x^{-½}$ |
| Exponential | $e^x$ | $e^x$ |
| | $a^x$ | $\ln(a)\ a^x$ |

# Derivative Rules

| | | |
|---|---|---|
| Logarithms | $\ln(x)$ | $1/x$ |
| | $\log_a(x)$ | $1 / (x \ln(a))$ |
| Trigonometry (x is in radians) | $\sin(x)$ | $\cos(x)$ |
| | $\cos(x)$ | $-\sin(x)$ |
| | $\tan(x)$ | $\sec^2(x)$ |
| Inverse Trigonometry | $\sin^{-1}(x)$ | $1/\sqrt{(1-x^2)}$ |
| | $\cos^{-1}(x)$ | $-1/\sqrt{(1-x^2)}$ |
| | $\tan^{-1}(x)$ | $1/(1+x^2)$ |

# Derivative Rules

| Rules | Function | Derivative |
|---|---|---|
| Multiplication by constant | $cf$ | $cf'$ |
| Power Rule | $x^n$ | $nx^{n-1}$ |
| Sum Rule | $f + g$ | $f' + g'$ |
| Difference Rule | $f - g$ | $f' - g'$ |
| Product Rule | $fg$ | $f\,g' + f'\,g$ |
| Quotient Rule | $f/g$ | $\frac{f'\,g - g'\,f}{g^2}$ |
| Reciprocal Rule | $1/f$ | $-f'/f^2$ |

# Derivative Rules

| | | |
|---|---|---|
| Chain Rule (as "Composition of Functions") | $f \circ g$ | $(f' \circ g) \times g'$ |
| Chain Rule (using ′ ) | $f(g(x))$ | $f'(g(x))g'(x)$ |
| Chain Rule (using $\frac{d}{dx}$ ) | $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$ | |